

AI Automation with n8n

Lesson 1: Introduction to n8n

Introduction to n8n – Study Notes

Key Concepts

| # | Concept | Why It Matters |

|---|-----|-----|

| 1 | **What n8n Is** – an open source, node based workflow automation tool that can connect APIs, databases, files, and services without writing code. | Gives you a free, self hostable alternative to SaaS automators (Zapier, Make). |

| 2 | **UI Overview** – the main panels (Sidebar, Canvas, Node Settings, Execution Log) and how to navigate them. | Knowing the layout lets you build, test, and debug workflows efficiently. |

| 3 | **Nodes & Connections** – each node represents an action/trigger; arrows define data flow. | The building blocks of every automation. |

| 4 | **Creating Your First Workflow** – adding a trigger, an action, configuring credentials, and running the workflow. | Hands on experience that cements the above concepts. |

| 5 | **Execution Modes** – Manual run, Triggered run, and Schedule (cron) run. | Determines when and how your workflow fires. |

Summary

n8n (pronounced “n-eight-n”) is a **free, open source workflow automation platform** that lets you stitch together APIs, databases, files, and cloud services using a visual, node based interface. Unlike many proprietary tools, n8n can be self hosted on your own server, Docker container, or the n8n.cloud service, giving you full control over data privacy and the ability to extend functionality with custom JavaScript code.

The UI is divided into four primary areas: the **Sidebar** (where you pick nodes and access settings), the **Canvas** (the workspace where you drag and drop nodes and draw connections), the **Node Settings Panel** (on the right, where you configure each node’s parameters and credentials), and the **Execution Log** (bottom, showing run history and debugging info). Building a workflow starts with a **trigger node** (e.g., “Webhook” or “Cron”) that initiates the flow, followed by one or more **action nodes** (e.g., “HTTP Request”, “Google Sheets”, “Send Email”). After configuring any required credentials, you can **run the workflow manually** to test it, then enable it to run automatically based on the trigger type.

By the end of this lesson you should be able to launch n8n, navigate the UI, create a simple “Hello World” workflow that receives a webhook request and replies with a message, and understand the basics of how data moves through nodes.

Important Points

- **Open source & self hostable** – you can run n8n locally, on a VPS, or via Docker; no vendor lock in.

- **Node based design** – every integration is a node; nodes are connected with arrows that represent the flow of JSON data.
- **Triggers vs. Actions**
 - **Trigger nodes** start a workflow (Webhook, Cron, Polling, etc.).
 - **Action nodes** perform tasks (HTTP request, DB query, email, etc.).
 - **Credentials** are stored centrally (Settings ! Credentials) and referenced in node configs – never hard code API keys in the workflow.
 - **Execution Modes**
 - **Manual** – click “Execute Workflow” to test.
 - **Triggered** – runs automatically when the trigger fires.
 - **Scheduled** – runs on a cron schedule.
 - **Debugging** – the Execution Log shows each node’s input and output; you can pause at any node to inspect data.
 - **Extensibility** – use the “Function” node for custom JavaScript, or create your own node via the n8n community or by writing a custom node package.

Examples

Example 1 – Simple Webhook ! HTTP Response

1. **Add a “Webhook” trigger** (Sidebar ! Trigger ! Webhook).
2. Set **Path** to `hello`. n8n will generate a URL like `https://your n8n-instance/webhook/hello`.
3. **Add a “Respond to Webhook” node** (Sidebar ! Response ! Respond to Webhook).
4. In the Respond node, set **Response Body** to:

```
```json
{
 "message": "Hello from n8n!",
 "received": {{$json["body"]}}
}
```
```

5. Connect the Webhook node ! Respond node.
6. Click **Execute Workflow** ! copy the webhook URL and send a POST request (e.g., via `curl` or Postman).
7. The response will contain the JSON defined above, confirming the workflow works.

***Explanation*:** The webhook node receives an incoming HTTP request, passes the request payload as JSON to the next node, and the Respond node sends a custom JSON reply. This demonstrates the basic trigger action pattern and data flow.

Example 2 – Cron Trigger ! Google Sheets Append

1. **Add a “Cron” trigger** (Sidebar ! Trigger ! Cron). Set it to run **Every 5 minutes**.
2. **Add a “Google Sheets” node** (Sidebar ! Google Sheets ! Append).
3. Authenticate with your Google account (Credentials ! Google Sheets OAuth2).
4. Choose the spreadsheet and sheet, then map fields:
 - **Timestamp**: `{{\$now}}`
 - **Message**: “Automated entry”

5. Connect Cron !' Google Sheets.

6. **Activate** the workflow (toggle “Active”).

Explanation: Every five minutes the Cron node fires, passing the current time to the Google Sheets node, which appends a new row. This shows scheduled automation and using built in expressions (`{{ $now }}`).

Quick Review

1. **What does n8n stand for, and why is it considered “fair code”?**

2. **Name the four main UI sections of the n8n editor.**

3. **What is the difference between a Trigger node and an Action node?**

4. **How do you securely store API keys for use in multiple nodes?**

5. **Describe how you would debug a workflow that isn't producing the expected output.**

Further Reading

- **n8n Documentation** – deep dive into node reference, credential types, and expression syntax.
- **Creating Custom Nodes** – how to extend n8n with your own JavaScript/TypeScript nodes.
- **Workflow Best Practices** – naming conventions, error handling with “Error Trigger”, and using “Set”/“Function” nodes for data transformation.
- **Self Hosting Options** – Docker, Kubernetes, and serverless deployments.
- **Comparing n8n with Zapier/Make** – strengths, limitations, and migration strategies.

